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| 9(a) | <p>any <b>eight</b> from:</p> <ol style="list-style-type: none"> <li>1 vary / different / range of, glucose / fructose, concentrations ;</li> <li>2 control / same, volume of yeast (suspension) ;</li> <li>3 control / same, <u>volume</u> of glucose (solution) ;</li> <li>4 control / same, volume / concentration, of indicator (solution) ;</li> <li>5 control temperature (between 20–45°C) ;</li> <li>6 oil layer to exclude, air / oxygen ;</li> <li>7 indicator turns from blue (start / oxidised) to colourless (end / reduced) ;</li> <li>8 due to, hydrogen atoms / <math>H^+</math> / electrons, from, glycolysis / dehydrogenation / respiration ;</li> <li>9 time how long it takes for colour change <b>OR</b> fix time interval and measure with colorimeter ;</li> <li>10 repeat experiment ;</li> <li>11 calculate mean(s) ;</li> <li>12 graph with glucose concentration (on x axis) and, time taken / % absorbance / % transmission / rate (on y) ;</li> <li>13 rate is <math>1/t</math> ;</li> <li>14 AVP ;</li> </ol> | 8 |
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| Question                    | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks                 |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------|-------------|-----------------------------|-------------------------|---------|-----------------|-----------------------|---------|---|--------------------------------------|-----------------|------------|-----------------------|-----------------------|------------------|-----------------------|-----------------------|--------------|------------|--------------|---|
| 9(b)                        | <p>any <b>seven</b> from:</p> <p><i>both</i></p> <ol style="list-style-type: none"> <li>1 occur in cytoplasm ;</li> <li>2 (only) involve glycolysis ;</li> <li>3 make, 2 (net) / small amount of, ATP ;</li> <li>4 regenerate NAD (from NADH) ;</li> </ol> <p><i>comparison</i></p> <table> <tr> <th></th><th>mammalian tissue</th><th>yeast cells</th></tr> <tr> <td>5 H acceptor / reduction of</td><td>pyruvate / pyruvic acid</td><td>ethanal</td></tr> <tr> <td>6 (end) product</td><td>lactate / lactic acid</td><td>ethanol</td></tr> <tr> <td>7</td><td>no, CO<sub>2</sub>/decarboxylation</td><td>CO<sub>2</sub></td></tr> <tr> <td>8 (enzyme)</td><td>lactate dehydrogenase</td><td>ethanol dehydrogenase</td></tr> <tr> <td>9 steps / stages</td><td><u>one</u> / <u>1</u></td><td><u>two</u> / <u>2</u></td></tr> <tr> <td>10 (process)</td><td>reversible</td><td>irreversible</td></tr> </table> |                       | mammalian tissue | yeast cells | 5 H acceptor / reduction of | pyruvate / pyruvic acid | ethanal | 6 (end) product | lactate / lactic acid | ethanol | 7 | no, CO <sub>2</sub> /decarboxylation | CO <sub>2</sub> | 8 (enzyme) | lactate dehydrogenase | ethanol dehydrogenase | 9 steps / stages | <u>one</u> / <u>1</u> | <u>two</u> / <u>2</u> | 10 (process) | reversible | irreversible | 7 |
|                             | mammalian tissue                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | yeast cells           |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
| 5 H acceptor / reduction of | pyruvate / pyruvic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ethanal               |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
| 6 (end) product             | lactate / lactic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ethanol               |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
| 7                           | no, CO <sub>2</sub> /decarboxylation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO <sub>2</sub>       |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
| 8 (enzyme)                  | lactate dehydrogenase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ethanol dehydrogenase |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
| 9 steps / stages            | <u>one</u> / <u>1</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <u>two</u> / <u>2</u> |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |
| 10 (process)                | reversible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | irreversible          |                  |             |                             |                         |         |                 |                       |         |   |                                      |                 |            |                       |                       |                  |                       |                       |              |            |              |   |